

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 3: Lift & Drag Coefficients, Formulas & Polar Curves

Syllabus Source: CAE Oxford (pp. 73-108) | Pass Target: >= 90% AviationExam

1. THE AERODYNAMIC LIFT FORMULA

Total aerodynamic lift generated by a wing is governed by the universal lift formula:

- Lift Equation: $L = C_L \times \frac{1}{2} \rho V^2 \times S$, where C_L = Lift Coefficient (dimensionless); ρ = ambient air density (kg/m^3); V = True Airspeed (TAS, m/s); S = Wing surface area (m^2).
- Dynamic Pressure Component ($q = \frac{1}{2} \rho V^2$): Lift is directly proportional to dynamic pressure and wing planform area.
- Lift Coefficient (C_L): Represents airfoil lifting efficiency. Depends on: 1. Angle of attack (α); 2. Camber; 3. Airfoil shape; 4. Reynolds Number; 5. Mach number.
- C_L vs α Curve: Linear increase from zero-lift angle of attack up to $C_{L_{\text{max}}}$. For positively cambered airfoils, zero lift occurs at a NEGATIVE angle of attack (typically -2° to -4°). Symmetrical airfoils produce zero lift at EXACTLY 0° AOA.

2. THE DRAG FORMULA & DRAG COEFFICIENT (C_D)

Total aerodynamic drag resists aircraft forward motion through the air:

- Drag Equation: $D = C_D \times \frac{1}{2} \rho V^2 \times S$, where C_D = Drag Coefficient.
- Total Drag Decomposition: Total Drag = Parasite Drag (Profile Drag) + Induced Drag (Vortex Drag): $C_D = C_{Dp} + C_{Di}$.
- Lift-to-Drag Ratio (L/D): $L/D = C_L / C_D$. Measures aerodynamic efficiency. Maximum L/D ratio (L/D_{max}) occurs at the tangent from the origin to the polar curve, corresponding to the MINIMUM DRAG SPEED (V_{md})!

3. THE DRAG POLAR (LILIENTHAL POLAR CURVE)

Key Polar Feature	Aerodynamic Meaning & Optimum Point	Flight Application & Performance Target
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Origin Tangent Point	Maximum Lift-to-Drag Ratio: (C_L / C_D) _{max} .	Minimum Drag Speed (V_{md}). Yields: 1. Maximum Glide Range (unpowered); 2. Maximum Endurance for jet aircraft; 3. Maximum Range for piston-prop aircraft.
Peak of the Curve	Maximum Lift Coefficient ($C_{L_{\text{max}}}$).	Minimum Flying Airspeed (Stall Speed V_S). Corresponds to stalling angle of attack (α_{crit} , $\sim 15\text{-}18^\circ$).
Maximum ($C_L^{1/2} / C_D$)	Optimum speed for jet aircraft cruising range.	Long-range cruise speed for jet airliners (typically $1.32 \times V_{md}$).
Maximum ($C_L^{3/2} / C_D$)	Minimum rate of sink / Minimum power required.	Maximum endurance for propeller aircraft; minimum sink speed for gliders (typically $0.76 \times V_{md}$).

[!] EXAM TRAP (AVIATIONEXAM) - Glide Performance vs Aircraft Gross Mass Trap

- QUESTION: How does increasing aircraft gross mass affect the maximum glide distance in power-off flight? - ANSWER: Gross mass HAS ZERO EFFECT ON MAXIMUM GLIDE DISTANCE! - Maximum glide ratio equals $(L/D)_{max}$, which is purely aerodynamic. - Effect of Heavier Weight: The aircraft glides at a HIGHER TRUE AIRSPEED and has a HIGHER RATE OF DESCENT, covering the exact same ground distance in a shorter time!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Weight Variation in Gliding Flight**

Problem: An aircraft glides at 100 kt with engines idle. If gross weight increases by 21%, what happens to glide range and best glide speed? Step 1: Glide range depends exclusively on L/D_{max} . Clean aerodynamics are unchanged, so glide range in still air DOES NOT CHANGE. Step 2: Best glide speed increases with the square root of weight ratio: $V_{new} = 100 \times \sqrt{1.21} = 100 \times 1.10 = 110$ kt. Conclusion: The heavier aircraft glides at 110 kt vs 100 kt, reaches the ground in less time, but lands at the EXACT SAME distance.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: If an aircraft doubles its airspeed ($2 \times V$) while maintaining constant altitude and angle of attack, by what factor does lift increase?

- [A]: By a factor of 2
- [B]: By a factor of 4 (quadruples)
- [C]: By a factor of 8
- [D]: Lift remains constant

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Lift formula: $L = C_L \times \frac{1}{2} \rho V^2 \times S$. Because lift is proportional to the square of velocity (V^2), doubling airspeed produces $(2)^2 = 4$ times more lift.

Q2: At what angle of attack does a SYMMETRICAL airfoil generate zero aerodynamic lift?

- [A]: At -4° angle of attack
- [B]: At exactly ZERO degrees (0°) angle of attack
- [C]: At $+2^\circ$ angle of attack
- [D]: At stalling angle

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Symmetrical airfoils have identical upper and lower contours with zero camber. Airflow acceleration is identical on both surfaces at 0° AOA, producing zero net lift. (Positively cambered airfoils require negative AOA for zero lift).

Q3: What operational flight condition corresponds to the point of Maximum Lift-to-Drag Ratio $((L/D)_{max})$?

- [A]: Maximum airspeed
- [B]: Minimum Total Drag Speed (V_{md}) and Maximum Glide Range
- [C]: Maximum stall speed
- [D]: Minimum climb angle

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At $(L/D)_{\max}$, aerodynamic efficiency is optimized. Total drag is at its absolute minimum (V_{md}), providing the shallowest glide angle and maximum unpowered glide distance.